

KAJAANI, Finland

WMO: 028970

Lat: 64.283N

Lon: 27.673E

Elev: 134

StdP: 99.73

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.3	-27.0	-33.6	0.2	-30.3	-29.9	0.2	-26.9	9.9	-2.0	8.7	-2.9	0.3	80	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	25.7	17.8	23.8	16.9	22.0	15.9	19.2	23.5	17.9	21.8	16.8	20.6	3.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.7	21.2	16.2	11.7	19.9	15.1	10.9	18.8	55.0	23.7	51.0	21.6	47.7	20.6	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.3	DB	-32.9	27.9	3.6	2.4	-35.5	29.6	-37.6	31.0	-39.6	32.4	-42.2	34.2	(4)
(5)			WB	-33.0	20.6	3.6	1.8	-35.6	21.8	-37.6	22.9	-39.6	23.9	-42.2	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-10.3	-10.4	-5.7	1.0	7.3	12.7	15.9	13.7	8.9	2.7	-2.6	-6.9	(6)
(7)		DBStd	10.54	8.03	8.02	5.84	4.41	4.53	3.69	3.30	3.35	3.51	4.10	5.19	7.43	(7)
(8)		HDD10.0	3274	630	570	486	269	108	14	1	7	62	226	377	523	(8)
(9)		HDD18.3	5882	888	803	744	519	342	172	89	147	284	483	627	782	(9)
(10)		CDD10.0	458	0	0	0	1	25	96	186	122	28	1	0	0	(10)
(11)		CDD18.3	24	0	0	0	0	1	4	14	4	0	0	0	0	(11)
(12)		CDH23.3	201	0	0	0	0	13	41	114	32	0	0	0	0	(12)
(13)		CDH26.7	24	0	0	0	0	1	4	17	3	0	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.4	2.6	2.9	2.8	3.1	3.0	2.6	2.6	2.8	3.0	3.0	2.7	(14)
(15)	Precipitation	PrecAvg	559	34	27	27	28	44	57	69	82	59	47	44	36	(15)
(16)		PrecMax	723	62	62	67	66	111	134	169	152	108	100	86	71	(16)
(17)		PrecMin	418	14	7	4	9	0	12	17	16	10	0	10	10	(17)
(18)		PrecStd	77	13	13	13	15	25	26	35	33	25	24	19	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.1	4.1	7.4	16.7	25.2	27.2	28.6	26.6	20.4	12.6	8.6	5.0	(19)
(20)			MCWB	2.2	2.4	3.5	10.0	14.3	17.2	19.4	18.8	15.4	10.7	7.8	4.8	(20)
(21)		2%	DB	1.5	2.4	4.8	12.7	21.8	24.2	26.2	23.9	17.8	11.0	6.7	3.2	(21)
(22)			MCWB	1.0	1.4	2.0	7.2	13.3	16.3	18.6	17.7	14.0	9.7	6.0	2.7	(22)
(23)		5%	DB	0.7	1.0	3.0	10.1	18.4	21.9	24.2	21.9	15.9	9.7	4.9	2.0	(23)
(24)			MCWB	0.3	0.4	1.0	5.4	11.3	15.1	17.6	16.6	12.8	8.7	4.4	1.5	(24)
(25)		10%	DB	-0.2	-0.1	1.7	7.8	15.7	19.9	22.5	20.0	14.3	8.4	3.5	1.0	(25)
(26)			MCWB	-0.5	-0.6	0.3	4.3	9.9	14.0	16.5	15.7	12.1	7.5	3.0	0.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.7	4.0	10.6	15.6	19.2	21.2	20.2	16.0	11.3	7.8	4.6	(27)
(28)			MCDWB	2.8	3.7	6.4	14.8	21.8	23.6	26.0	24.9	19.0	12.2	8.6	4.8	(28)
(29)		2%	WB	1.0	1.4	2.4	7.8	13.9	17.6	19.7	18.2	14.5	9.9	5.9	2.8	(29)
(30)			MCDWB	1.3	2.2	4.1	11.6	20.1	22.1	24.3	22.0	17.1	10.8	6.5	3.1	(30)
(31)		5%	WB	0.3	0.5	1.4	5.9	12.1	16.0	18.6	17.2	13.4	8.8	4.3	1.5	(31)
(32)			MCDWB	0.5	1.0	2.6	9.2	17.6	20.6	22.8	20.8	15.4	9.5	4.8	1.9	(32)
(33)		10%	WB	-0.7	-0.6	0.5	4.5	10.5	14.6	17.4	16.1	12.4	7.6	3.0	0.5	(33)
(34)			MCDWB	-0.3	-0.1	1.5	7.4	14.8	19.0	21.2	19.7	14.2	8.3	3.4	0.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.8	7.8	8.9	8.9	9.9	9.3	9.3	8.9	7.2	4.7	4.5	5.9	(35)
(36)			MCDBR	4.9	5.5	8.3	12.3	15.3	12.6	12.2	11.7	9.4	5.5	4.4	3.8	(36)
(37)			MCWBR	4.8	5.1	6.5	7.8	8.1	6.7	6.0	6.7	6.3	4.5	4.3	3.7	(37)
(38)		5% WB	MCDBR	5.0	5.3	6.7	11.4	13.6	10.8	10.4	10.3	8.5	5.1	4.3	3.7	(38)
(39)			MCWBR	4.9	5.0	5.4	7.6	7.8	6.1	5.7	6.2	5.9	4.5	4.3	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.211	0.276	0.305	0.343	0.343	0.344	0.368	0.348	0.325	0.302	0.359	0.205	(40)	
(41)		taud	1.956	2.111	2.167	2.284	2.460	2.479	2.432	2.472	2.524	2.383	2.643	1.967	(41)	
(42)		Ebn at Noon	339	619	770	829	869	874	838	823	771	609	318	126	(42)	
(43)		Edh at Noon	27	62	90	102	95	95	97	85	66	49	24	11	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.14	0.66	2.08	3.60	4.71	5.24	4.96	3.70	2.05	0.75	0.18	0.05	(44)	
(45)		RadStd	0.02	0.14	0.21	0.39	0.51	0.38	0.50	0.36	0.21	0.12	0.04	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-182	N/A	N/A
(47)	Regional (31 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page